Kanoon-Bridge

A Multi-Layer Information Retrieval System for India's New Criminal Codes

Demo guide

Gaurav Rai (2410110133) Shaurya Sharma (2410110317) Sharanya Gupta (2410110315)

https://github.com/gauravrai1704/Kanoon-bridge

What this guide is. A script for a 10–12 minute live demo: what to type, what appears (every screenshot is the real web interface, taken on the IL-PCSR build), what to say, and which novelty or experiment each step shows. Every question below was run and checked on our index; the *avoid* list (§7) names the questions that are weak today.

Run sheet

Time	Scene	What it shows	Where
0:00	Pitch	the problem: 1 July 2024, colliding section numbers	§2
1:00	1	the interface and its controls	web
1:30	2	the incident date decides the code (IPC 302 vs BNS 103)	web
2:30	3	an old IPC number answered under the new law (420 → BNS 318)	web
3:00	4	a colliding number: "section 302" in 2023 vs 2025	web
4:00	5	Hinglish, Hindi and typos	web
5:00	6	under the hood: research agent, verifier, every step	web
6:00	7	statutory drift: the "What changed" page (IPC 124A → BNS 152)	web
7:00	8	binding courts for the State, overruled precedent	web
8:00	9	honest abstention; Boolean search	web
8:30	10	a local LLM, its claims checked and corrected (Ollama)	web / figures
9:30	Tests	211 unit tests live; the evaluation in one table	terminal, §4
10:30	Close	the novelties in one minute	§5

1 Before the demo (10 minutes earlier)

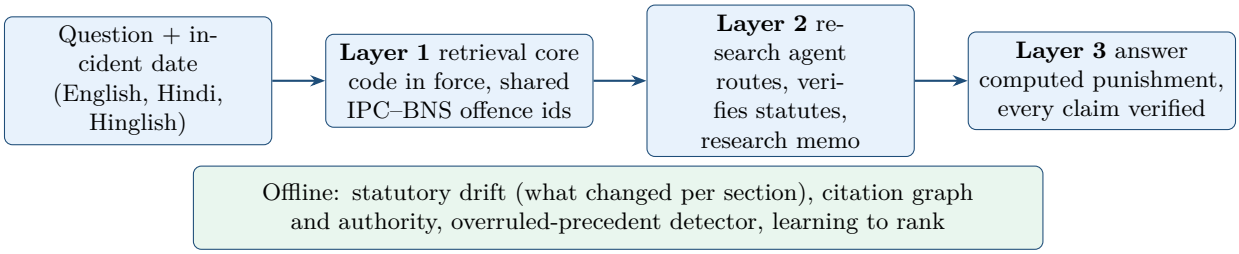
1. **Start the web interface** (the index is already built on the demo laptop):

```
cd Kanoon-bridge && source .venv/bin/activate      # Windows: .venv\Scripts\activate
make web                                           # Windows: python app/web_server.py
```

   Open `http://localhost:8000`. The left panel says how many statutes and judgments are indexed (1,995 and 3,183) when it is ready (about 10 seconds).
2. **For scene 10 only**, start Ollama in a second terminal: `OLLAMA_CONTEXT_LENGTH=8192 ollama serve` (model `llama3.2:3b` already pulled). "Answer by" then lists it.
3. **Warm up**: ask "punishment for murder" once, then "New question". The first search loads the index into memory.
4. **Projector**: browser zoom 110–125%; switch the research agent **on** and keep it on for every scene.
5. **Backup**: keep this PDF open. If anything fails live, show the screenshot of that scene.

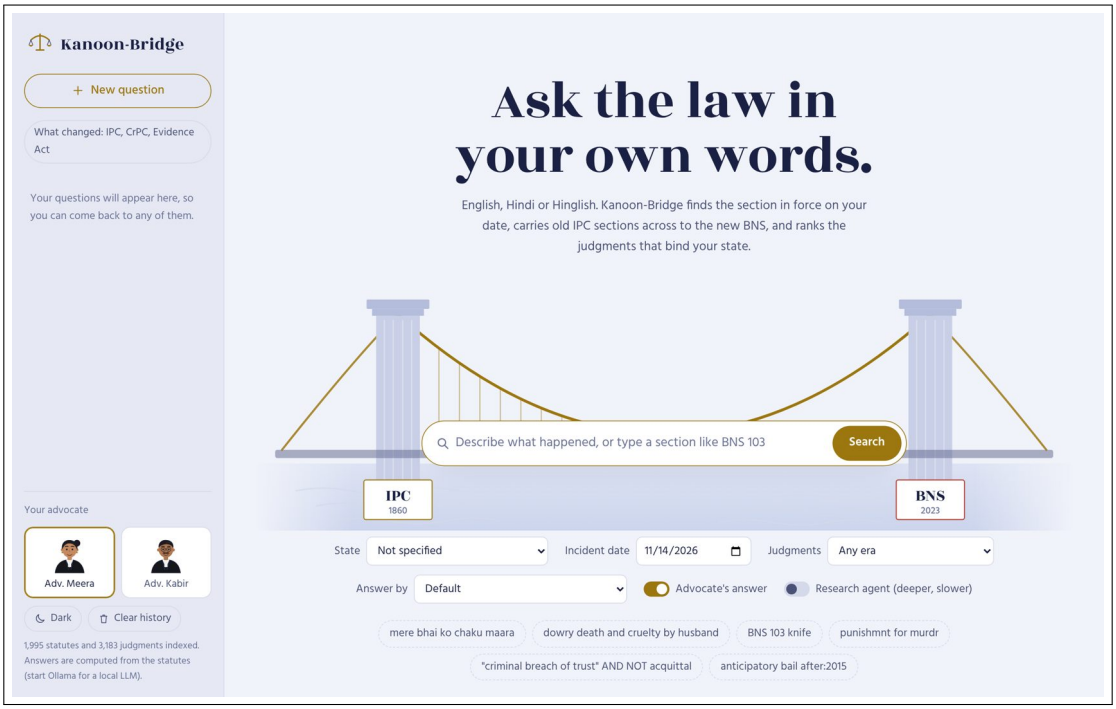
2 The 60-second pitch

Say: "On 1 July 2024 India replaced its three criminal codes in one day: the IPC became the BNS, the CrPC the BNSS, the Evidence Act the BSA. Almost every section number changed, and some numbers now mean different offences: section 302 is murder in the IPC but hurting religious feelings in the BNS. Which law applies depends on the date of the incident, and courts cite both codes: in 2025 only 8–29% of High Court judgments a month cite the BNS. Search engines and chatbots that ignore this give confident, wrong answers: a small local LLM cites the old code for a third of post-2024 questions. Kanoon-Bridge is a three-layer IR system that gets this right: a version-aware search core, a research agent that verifies what it finds, and an answer layer that computes and checks every claim."



3 Live demo, scene by scene

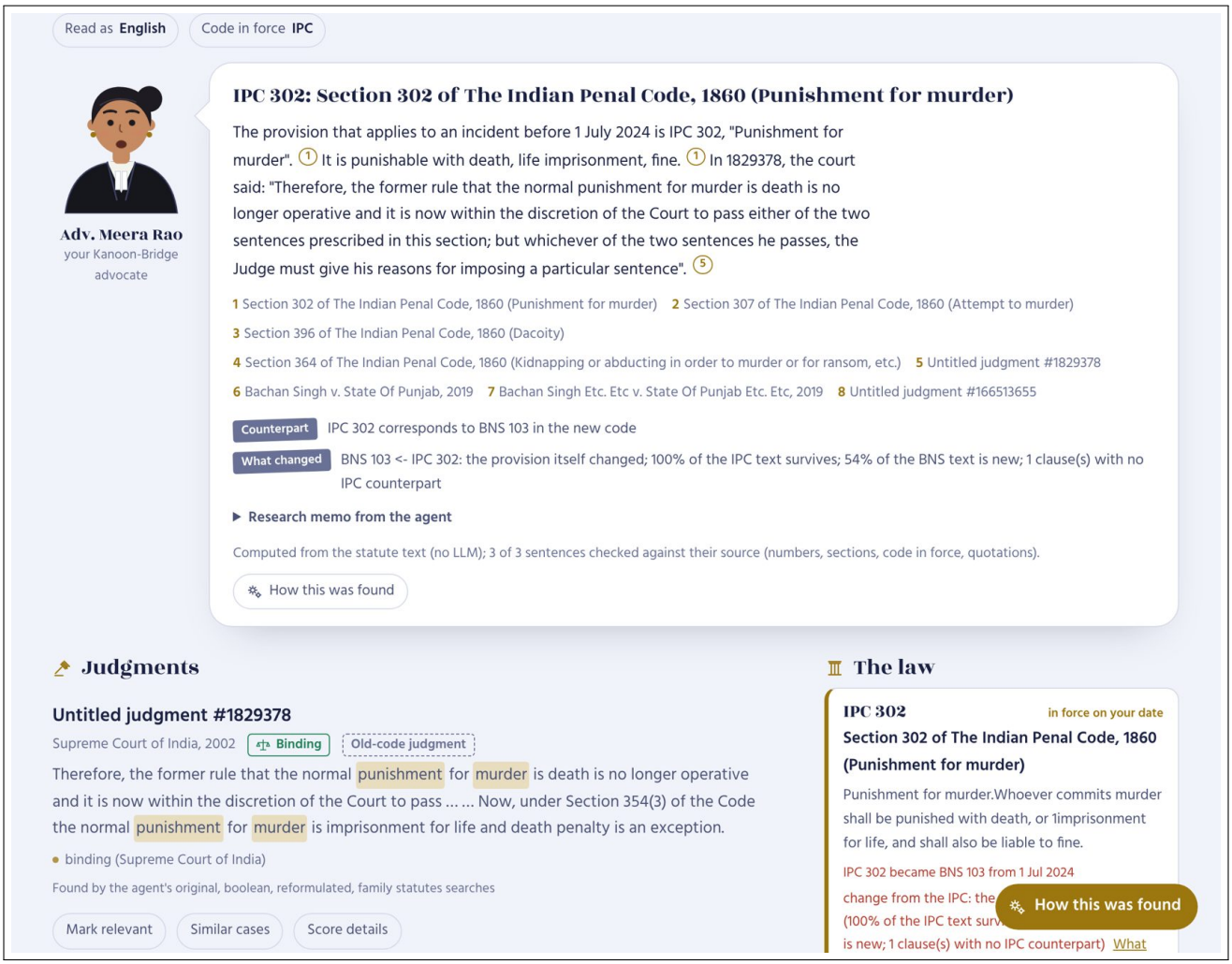
Scene 1 – The interface

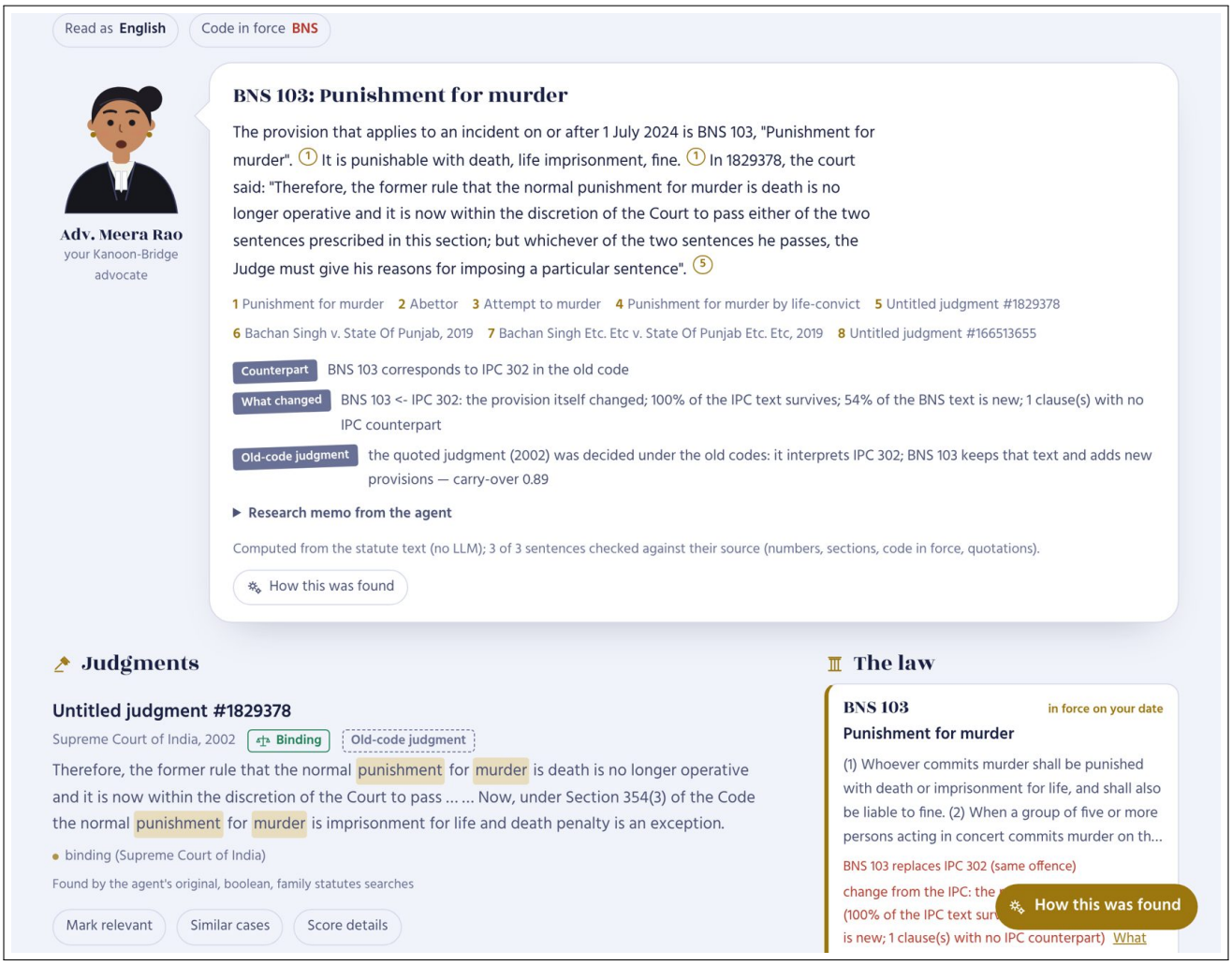


Point at: the question box (“describe what happened, or type a section”); **State** (decides which courts bind); **Incident date** (decides the code); **Judgments** (any, new codes, old codes); **Answer by** (computed, or a local LLM); the switches **Advocate’s answer** (layer 3) and **Research agent** (layer 2); the example chips; the link **What changed** (top left).

Scene 2 – The date decides the code

Type `punishment for murder`, date `2023-03-01`, search. Then change only the date to `2025-01-03` and search again.


Read as English
Code in force IPC
Adv. Meera Rao
your Kanoon-Bridge advocate
IPC 302: Section 302 of The Indian Penal Code, 1860 (Punishment for murder)
The provision that applies to an incident before 1 July 2024 is IPC 302, "Punishment for murder". 1 It is punishable with death, life imprisonment, fine. 1 In 1829378, the court said: "Therefore, the former rule that the normal punishment for murder is death is no longer operative and it is now within the discretion of the Court to pass either of the two sentences prescribed in this section; but whichever of the two sentences he passes, the Judge must give his reasons for imposing a particular sentence". 5
1 Section 302 of The Indian Penal Code, 1860 (Punishment for murder) 2 Section 307 of The Indian Penal Code, 1860 (Attempt to murder)
3 Section 396 of The Indian Penal Code, 1860 (Dacoity)
4 Section 364 of The Indian Penal Code, 1860 (Kidnapping or abducting in order to murder or for ransom, etc.) 5 Untitled judgment #1829378
6 Bachan Singh v. State Of Punjab, 2019 7 Bachan Singh Etc. Etc v. State Of Punjab Etc. Etc, 2019 8 Untitled judgment #166513655
Counterpart IPC 302 corresponds to BNS 103 in the new code
What changed BNS 103 <- IPC 302: the provision itself changed; 100% of the IPC text survives; 54% of the BNS text is new; 1 clause(s) with no IPC counterpart
Research memo from the agent
Computed from the statute text (no LLM); 3 of 3 sentences checked against their source (numbers, sections, code in force, quotations).
How this was found
Judgments
Untitled judgment #1829378
Supreme Court of India, 2002 Binding Old-code judgment
Therefore, the former rule that the normal punishment for murder is death is no longer operative and it is now within the discretion of the Court to pass ... ... Now, under Section 354(3) of the Code the normal punishment for murder is imprisonment for life and death penalty is an exception.
binding (Supreme Court of India)
Found by the agent's original, boolean, reformulated, family statutes searches
Mark relevant
Similar cases
Score details
The law
IPC 302
in force on your date
Section 302 of The Indian Penal Code, 1860 (Punishment for murder)
Punishment for murder.Whoever commits murder shall be punished with death, or 1imprisonment for life, and shall also be liable to fine.
IPC 302 became BNS 103 from 1 Jul 2024
change from the IPC: the
(100% of the IPC text surv
is new; 1 clause(s) with no IPC counterpart) What
How this was found

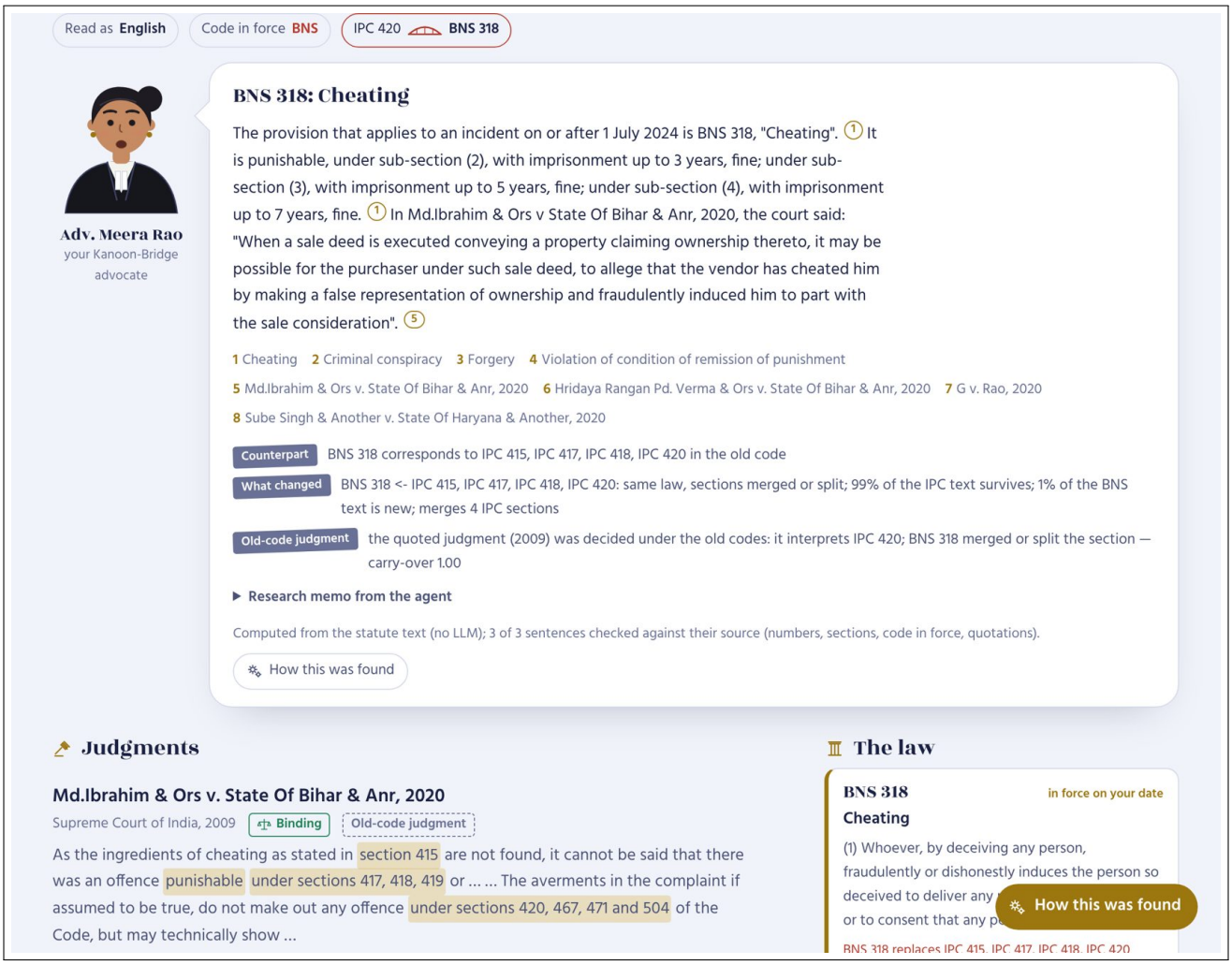


Say: "Same words, two dates. For 2023 it answers IPC 302; for 2025, BNS 103, the section in force. The punishment is computed from the statute text, not generated. The badges tell the lawyer what changed: BNS 103 corresponds to IPC 302; the IPC text survives completely, a new clause was added; and the quoted Supreme Court judgment was decided under the old code but still carries over (0.89)."

Novelty / evidence: date-based code filter + shared offence identifiers (E7: right code at rank 1 for 95.5% of questions vs 47.5% for BM25); computed punishments (E14: right punishment 86% vs 6% extractive).

Scene 3 – An old number, the new law

Type `punishment under section 420`, date `2025-01-03`.

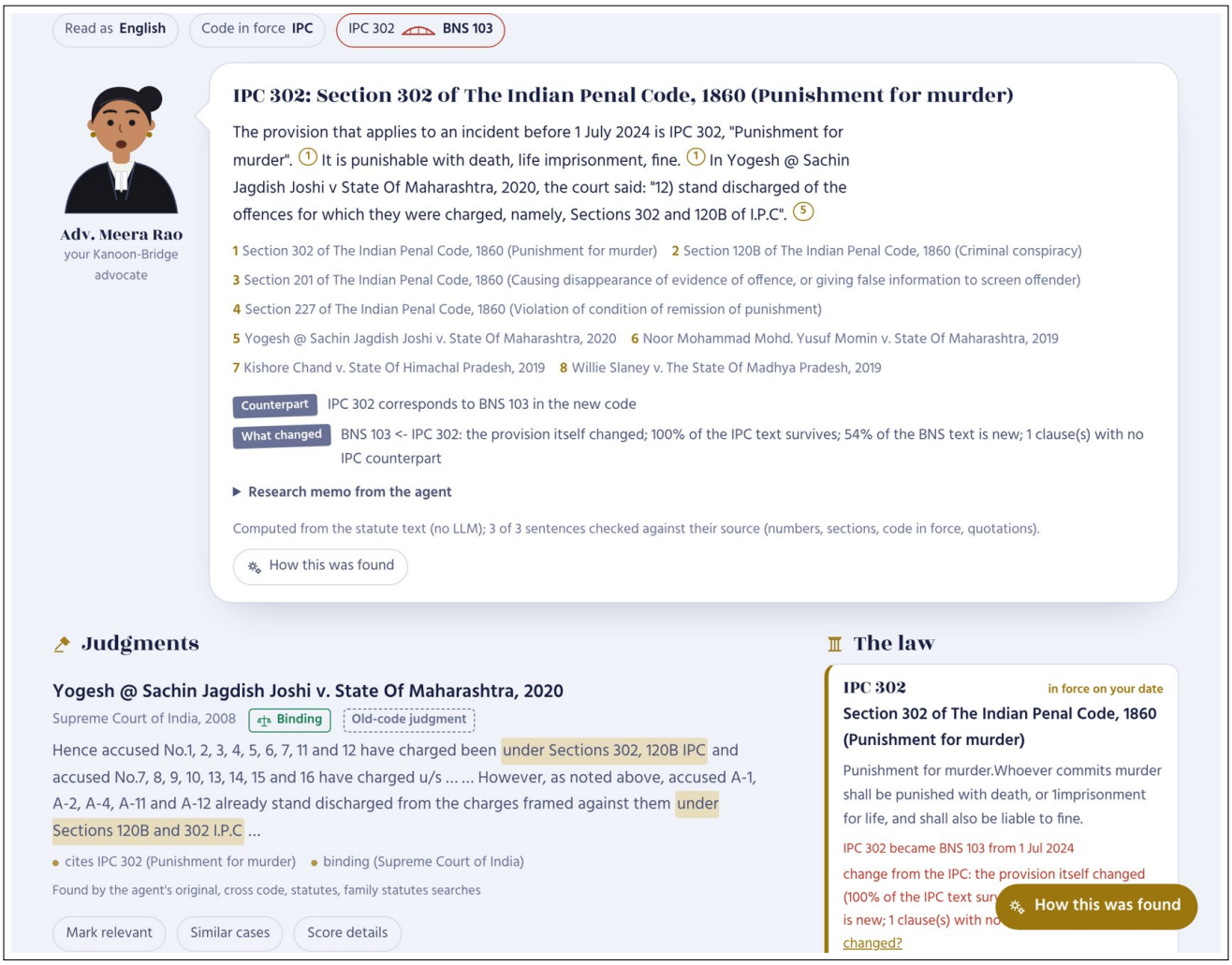


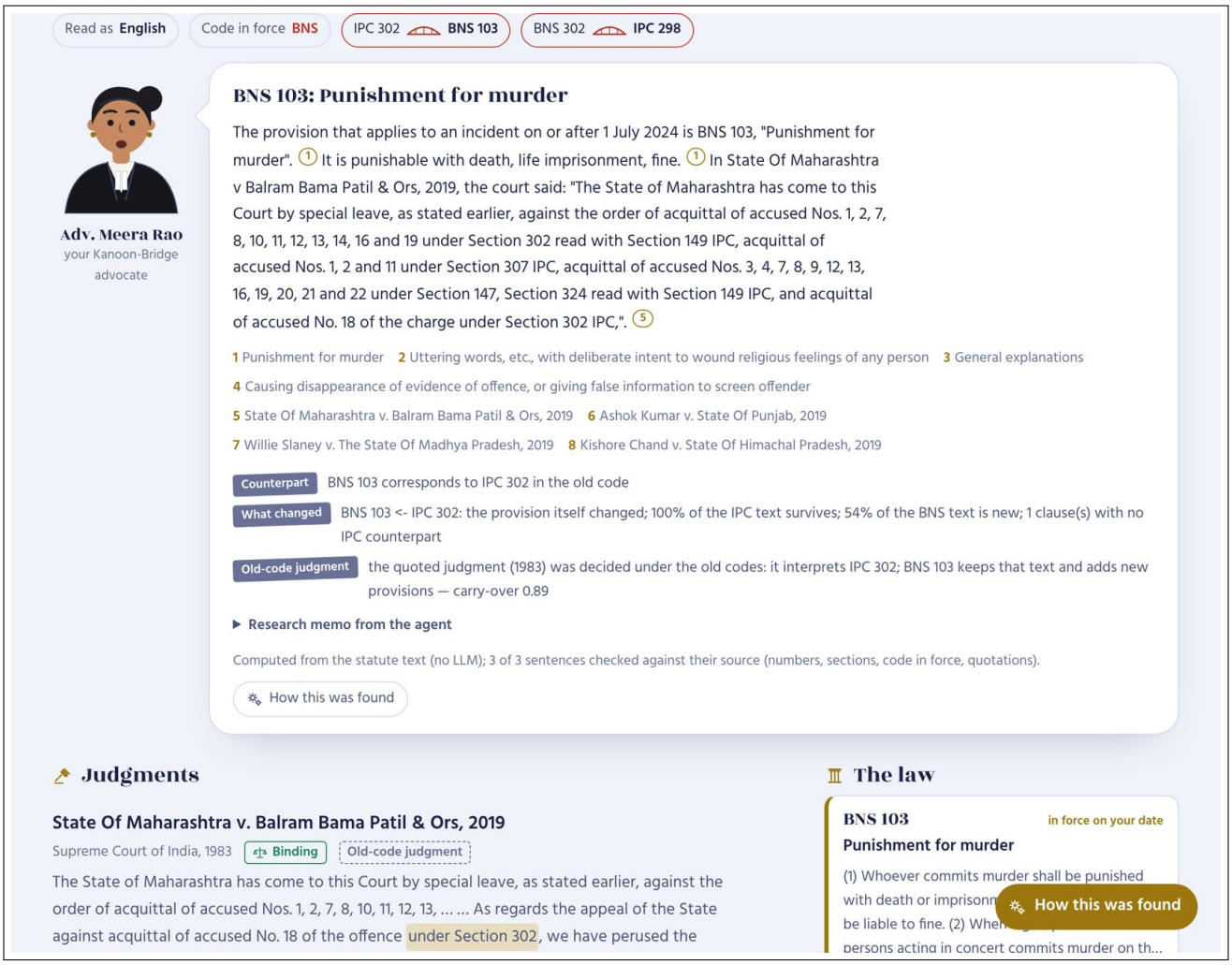
Say: "People still think in IPC numbers. 420 is cheating in the IPC; for a 2025 incident the system answers with BNS 318 and says it is the same law, merged from IPC 415, 417, 418 and 420."

Novelty / evidence: cross-version search: questions in new-code numbers find old-code precedents (E3: MAP 0.006 → 0.655; E8 procedure/evidence 0.005 → 0.654); beats BM25 *with* crosswalk expansion by 0.35 and 0.23 MAP (E23).

Scene 4 – A colliding number

Type `punishment under section 302`, date `2023-05-10`; then the same with `2025-02-10`.


Read as English
Code in force IPC
IPC 302 BNS 103
Adv. Meera Rao
your Kanoon-Bridge advocate
IPC 302: Section 302 of The Indian Penal Code, 1860 (Punishment for murder)
The provision that applies to an incident before 1 July 2024 is IPC 302, "Punishment for murder". 1 It is punishable with death, life imprisonment, fine. 1 In Yogesh @ Sachin Jagdish Joshi v State Of Maharashtra, 2020, the court said: "12) stand discharged of the offences for which they were charged, namely, Sections 302 and 120B of I.P.C". 5
1 Section 302 of The Indian Penal Code, 1860 (Punishment for murder) 2 Section 120B of The Indian Penal Code, 1860 (Criminal conspiracy)
3 Section 201 of The Indian Penal Code, 1860 (Causing disappearance of evidence of offence, or giving false information to screen offender)
4 Section 227 of The Indian Penal Code, 1860 (Violation of condition of remission of punishment)
5 Yogesh @ Sachin Jagdish Joshi v. State Of Maharashtra, 2020 6 Noor Mohammad Mohd. Yusuf Momin v. State Of Maharashtra, 2019
7 Kishore Chand v. State Of Himachal Pradesh, 2019 8 Willie Slaney v. The State Of Madhya Pradesh, 2019
Counterpart IPC 302 corresponds to BNS 103 in the new code
What changed BNS 103 <- IPC 302: the provision itself changed; 100% of the IPC text survives; 54% of the BNS text is new; 1 clause(s) with no IPC counterpart
Research memo from the agent
Computed from the statute text (no LLM); 3 of 3 sentences checked against their source (numbers, sections, code in force, quotations).
How this was found
Judgments
Yogesh @ Sachin Jagdish Joshi v. State Of Maharashtra, 2020
Supreme Court of India, 2008 Binding Old-code judgment
Hence accused No.1, 2, 3, 4, 5, 6, 7, 11 and 12 have charged been under Sections 302, 120B IPC and accused No.7, 8, 9, 10, 13, 14, 15 and 16 have charged u/s ... ... However, as noted above, accused A-1, A-2, A-4, A-11 and A-12 already stand discharged from the charges framed against them under Sections 120B and 302 I.P.C ...
cites IPC 302 (Punishment for murder) binding (Supreme Court of India)
Found by the agent's original, cross code, statutes, family statutes searches
Mark relevant Similar cases Score details
The law
IPC 302 in force on your date
Section 302 of The Indian Penal Code, 1860 (Punishment for murder)
Punishment for murder.Whoever commits murder shall be punished with death, or 1imprisonment for life, and shall also be liable to fine.
IPC 302 became BNS 103 from 1 Jul 2024
change from the IPC: the provision itself changed (100% of the IPC text sur
is new; 1 clause(s) with no
changed?
How this was found

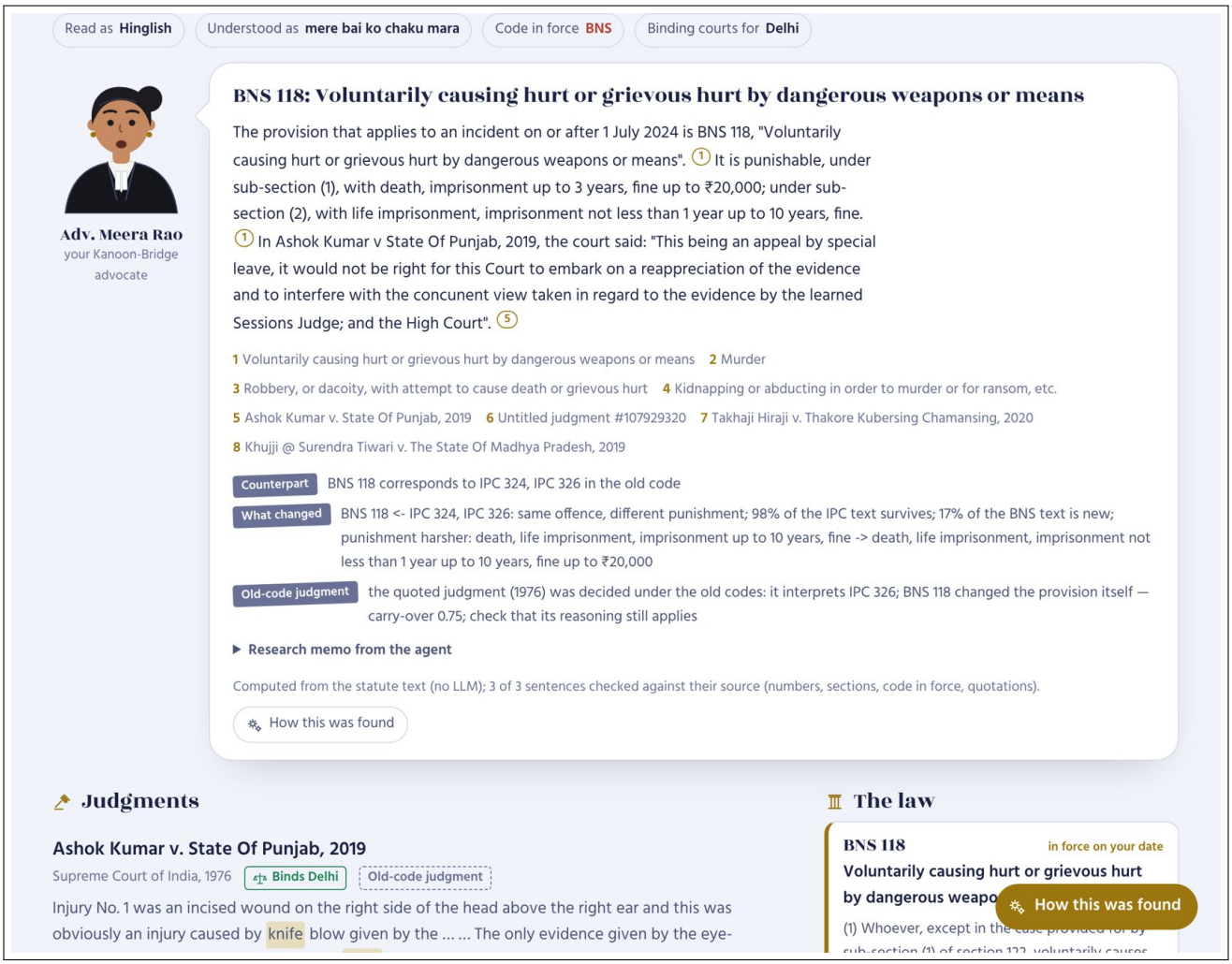


Say: "302 exists in both codes but means different things. For 2023: IPC 302, murder. For 2025, read literally, BNS 302 is hurting religious feelings, and the system says so and shows both readings (the IPC 302 → BNS 103 and BNS 302 ← IPC 298 chips at the top). A plain search engine mixes the two offences."

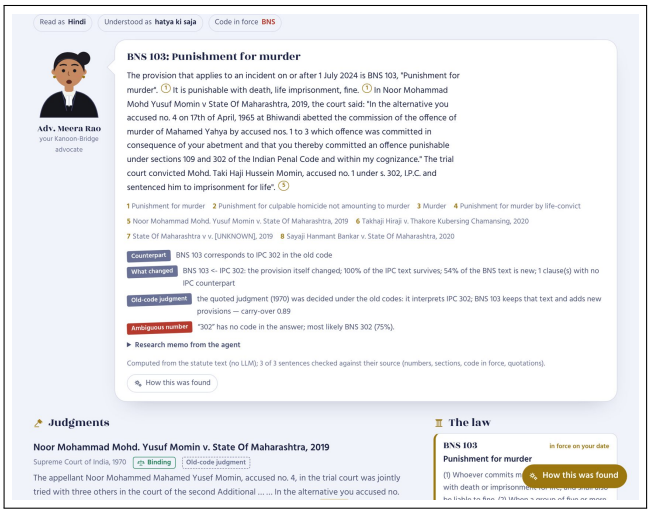
Novelty / evidence: collision resolution by date (E2: wrong-offence hits in the top 10 fall from 7.5% to 0%).

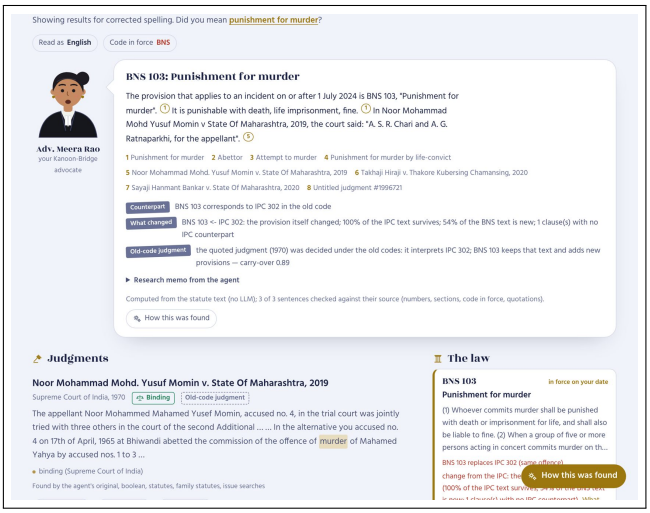
Scene 5 – Hinglish, Hindi and typos

Type `mere bhai ko chaku maara` ("my brother was stabbed with a knife"), State `Delhi`, date `2025-03-01`. Then `हत्या की सजा` (Hindi, "punishment for murder"), and `punishmnet for murdr`.



Say: "A victim's own words, in Hinglish. It is read as Hinglish, understood ('bai ko chaku mara'), and answered with BNS 118, grievous hurt by a dangerous weapon; judgments that bind in Delhi are marked."





Say: "Devanagari is transliterated and mapped through a legal lexicon: BNS 103. Misspellings are corrected against the index vocabulary ('Did you mean punishment for murder?')."

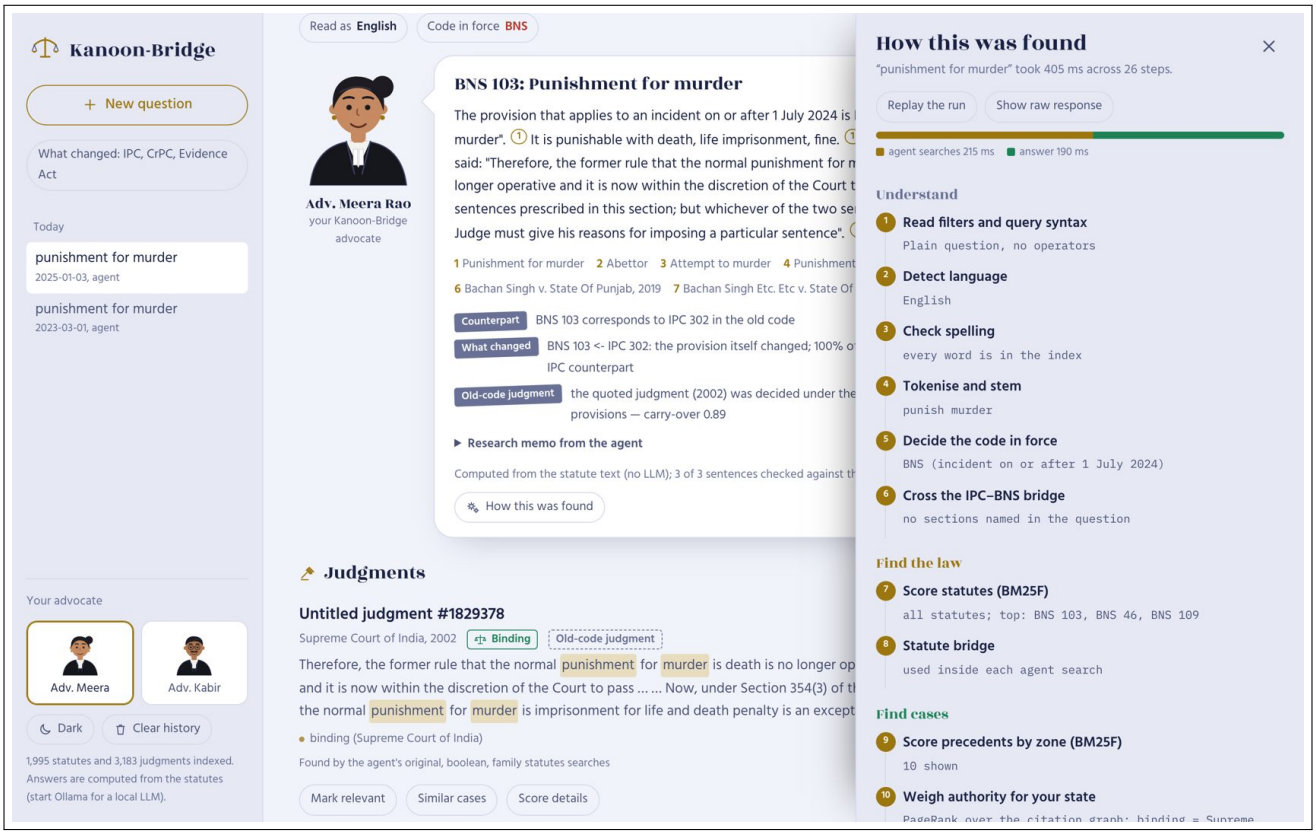
If asked about the red "Ambiguous number" badge on the Hindi answer: the verifier found a bare "302" inside the quoted judgment and flags that it has no code attached. That is the checker working, not an error in the answer.

Novelty / evidence: multilingual retrieval (E4: MAP 0.184 without the language steps → 0.461 over English, Hindi and Hinglish; the lexicon is the step that matters); spelling correction (MRR@10 0.830 → 0.905 on misspelled queries). The

Hindi/Hinglish components were built by Kashvi Tak for the mid-term and are used with her permission.

Scene 6 – Under the hood

After any search with the research agent on, click **How this was found**.

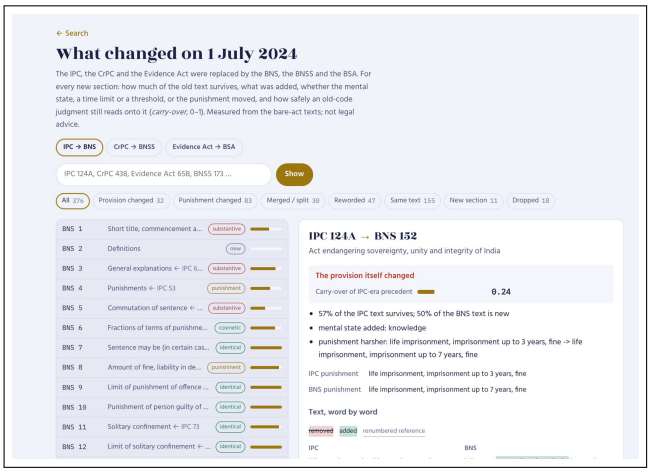


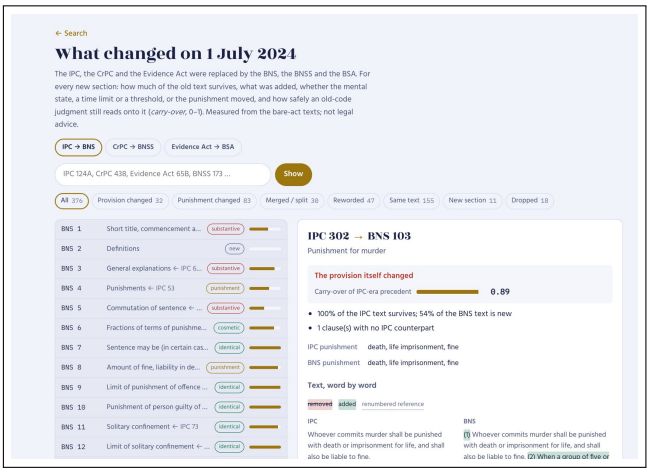
Say: "Every step is shown: language, spelling, the code in force, the IPC–BNS bridge, BM25F over judgment zones, the authority prior for your State, the agent's sub-searches and how they were fused, and the statute verifier. Nothing is a black box; 'Replay the run' replays it step by step."

Novelty / evidence: research agent v2 (routing, family refinement, statute verification): +0.044 MAP on procedure and evidence questions, +0.075 on Hindi/Hinglish (E17); on 995 real bail orders it doubles the core's MAP and nearly triples hit@1 (E19, $p_{\text{Holm}} < 0.001$).

Scene 7 – What changed: statutory drift

Click **What changed: IPC, CrPC, Evidence Act**, type `IPC 124A`, **Show**; then `IPC 302`.



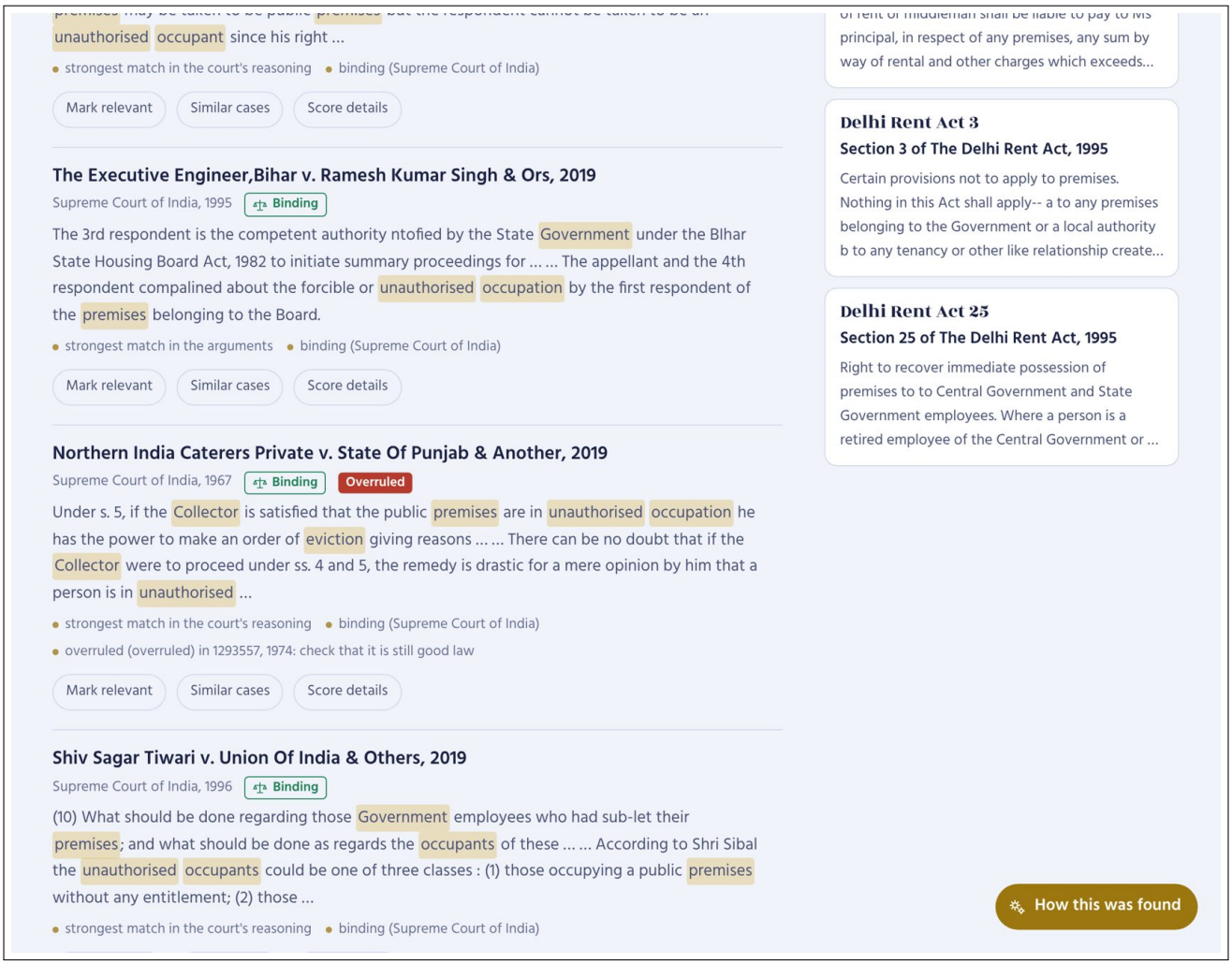


Say: "For every new section: how much of the old text survives, what was added, whether the mental state or the punishment changed, and how safely an old judgment still applies. Sedition, IPC 124A, became BNS 152: a new mental state, a harsher punishment, carry-over 0.24, so old sedition precedent is risky. Murder, IPC 302 to BNS 103: same offence, carry-over 0.89. The filters show the whole picture: 32 provisions changed in substance, 83 in punishment, 155 kept the same text."

Novelty / evidence: statutory drift detects changed provisions with F1 0.89 against an independent gold standard, where text similarity reaches 0.62 and the official crosswalk relation 0.21 (E11); the punishment parser is what matters (removing it drops F1 to 0.20).

Scene 8 – Binding courts and overruled precedent

Type `eviction of unauthorised occupant government premises collector` and scroll the judgments.

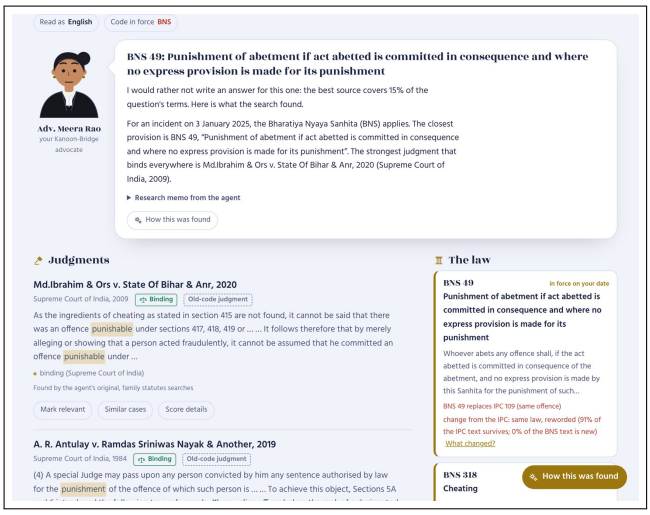


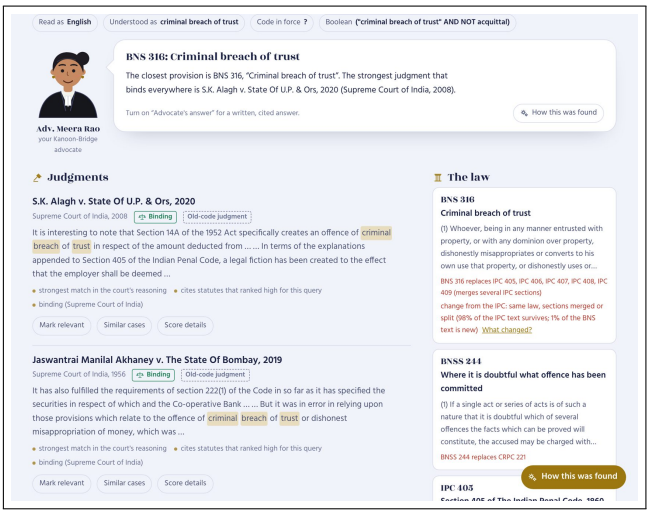
Say: "Each judgment says whether it binds in your State (Supreme Court everywhere, a High Court in its own State) and whether a later judgment overruled it: Northern India Caterers v. State of Punjab is marked overruled, with the citing judgment, found from the courts' own sentences."

Novelty / evidence: jurisdiction-aware ranking (E6: nDCG@10 0.366 → 0.657); overruled-precedent detector (E21): 5 judgments flagged overruled, all correct against the legal record.

Scene 9 – Honest abstention and Boolean search

Type `punishment for jaywalking`; then `"criminal breach of trust" AND NOT acquittal`.

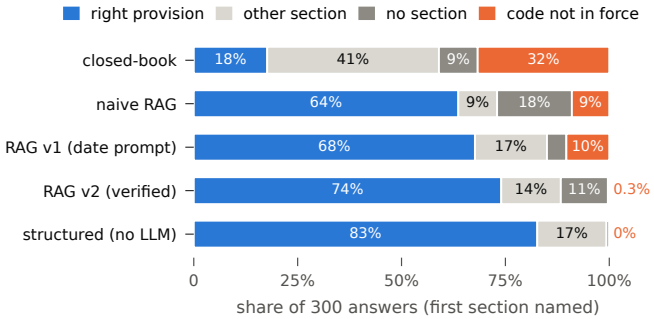


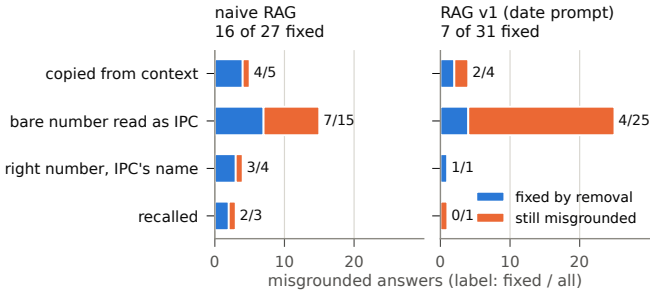


Say: "Jaywalking is not an offence in these codes, and the advocate says so instead of inventing one: the best source covers only 15% of the question. And for lawyers, full Boolean and phrase search with filters such as `after:2015` or `court:supreme_court`."

Scene 10 – A local LLM, checked claim by claim (Ollama)

With Ollama running, set **Answer by** to *Local model via Ollama (llama3.2:3b)*, then ask: `punishment for cheating`, date `2025-01-03`. The answer lists the claims checked and every correction the verifier made. If Ollama is not available, show these results instead:





Say: "We asked a local LLM 300 date-dependent questions (E20). Without retrieval it cites the code not in force 32% of the time, always the old code for new incidents. Telling it the date rule in the prompt does not help (10%). Our verification brings it to 0.3%. And when we removed the old-code passages from its context (E25), only some errors disappeared: most come from the model reading a bare number in the question as IPC. So checking the answer is necessary, not optional."

Novelty / evidence: first measurement of LLM temporal misgrounding at a code replacement, and of where it comes from.

4 Tests and evaluation

Live: the unit tests (about 10 seconds)

```
make test           # or: python -m pytest -q tests
.....................................................................  [100%]
211 passed in 7.72s
```

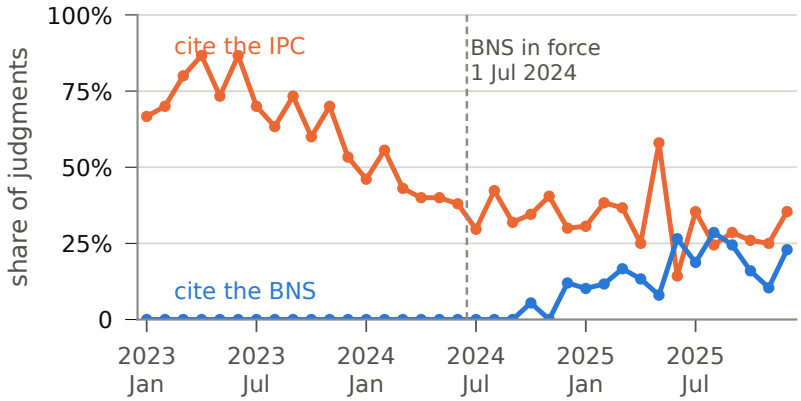
211 tests in 29 files cover the tokenizer and section parser, the IPC–BNS bridge and collision resolver, BM25F, n-grams, phonetics, spelling, Boolean search, learning to rank, the drift parser, the agent, claim verification, the local-LLM client and the context ablation.

The evaluation in one table

All results on the IL-PCSR corpus unless noted; paired randomization or exact McNemar tests, Holm-corrected; every number is produced by a script in the repository.

Set	Question	Baseline	Kanoon-Bridge
E1	pasted judgment → precedents / statutes (MAP)	0.220 / 0.149	**0.414 / 0.220**
E1 vs paper	IL-PCSR's published 3-gram baseline (MAP)	0.434 / 0.194	0.414 / **0.220**
E3 / E8	cases asked in new-code numbers (MAP)	0.006 / 0.005	**0.655 / 0.654**
E2	colliding numbers: wrong-offence hits in top 10	7.5%	**0%**
E7	right code at rank 1, incident before/after 1 July 2024	47.5%	**95.5%**
E4	English / Hindi / Hinglish needs (MAP)	0.184 (no language steps)	**0.461**
E6	jurisdiction: binding courts first (nDCG@10)	0.366	**0.657**
E11	which provisions changed (F1, held-out)	0.62 (text sim.)	**0.89**
E14	right punishment, 273 BNS questions	6% (extractive)	**86%**
E15	injected answer errors caught	17–100%	**82–100%**
E19	995 real bail orders → charged sections (MAP)	0.047	**0.100**
E20	LLM cites the code not in force	32% (closed-book)	**0.3%** (verified)
E24	"cases decided under the BNS" (MAP)	0.440 (BM25 + xw)	**0.595**

Comparison with a published paper: on IL-PCSR's own test split we beat its published lexical baseline on statutes (MAP 0.220 vs 0.194), are slightly below it on precedents (0.414 vs 0.434), and its trained models (0.502–0.544 MAP) remain ahead on precedents. Our contribution is the transition, which no earlier system measures.



The transition in 1,701 High Court judgments: the old code is still cited far more than the new.

5 The novelties in one minute

1. **The problem itself**: the first retrieval and RAG evaluation of a one-day replacement of three criminal codes with colliding section numbers, with eight new benchmarks.
2. **Version-aware retrieval**: shared offence identifiers across codes, inside the index, plus a date-based code filter and collision resolution.
3. **Statutory drift**: a per-provision measure of how far the law moved (text, mental state, thresholds, punishment) and a carry-over score for old precedent, validated against two independent sources.
4. **Verified research agent**: routes the question, chooses the code family, verifies the statutes it returns, and writes a research memo.
5. **Computed, claim-verified answers**: punishments computed from statute text; every claim checked for support, quantities, section identity, the code in force, negation and quotations.
6. **Judicial treatment**: overruled precedent detected from the judgments' own sentences.
7. **LLM temporal misgrounding**: measured on a local model (E20), and traced to its source by context ablation (E25).
8. **Era-aware retrieval**: "cases decided under the new code" as a separate intent (E24).
9. **Usable**: English, Hindi and Hinglish; typos; Boolean search; explanations for every result; runs fully offline with a local LLM.

6 Likely questions

Why not just map IPC numbers to BNS numbers with the government table?
The table maps sections, not meaning: 302 exists in both codes, sections were merged and split, and old precedent applies only where the law did not change. E23: naive crosswalk expansion is beaten by 0.23–0.35 MAP.

What is BM25F, and why zones?
BM25 with separate weights per field; judgments are split into facts, arguments, ratio and decision, so a match in the court's reasoning counts more. The formulas are in the report.

How do you know the drift labels are right?
Two independent gold sources (a retained/changed mapping and a study platform's notes) and a held-out split; the two sources agree at $\kappa = 0.75$ with each other.

Is the LLM answer trustworthy?
Not by itself: that is E20. Every sentence is verified against its source; failing claims are regenerated once, then repaired or removed, and every correction is shown.

What is new since the mid-term?
Drift (E11–E13), agent v2, answer layer v2, eCourts judgments, bail orders (E19), treatment (E21), KV reuse (E22), fair baselines (E23), era (E24), LLM study (E20, E25): about 22,600 lines of code. See the report's appendix.

What does not work?
Down-ranking old precedent by drift (it costs MAP), citation voting on real precedents, and training-free table-of-contents navigation; all reported as negative results.

Who did what?
Each member: prepare one or two sentences on your own components.

7 Avoid, and what to do if something breaks

Avoid live (current weak spots): theft, including "chori ki saza"; "anticipatory bail"; "punishment for sedition" typed as a question (show the What changed page instead); "dahej mrityu"; questions unrelated to criminal law (e.g. income tax); and the **Judgments: new codes** filter, which needs the eCourts judgments (not in the IL-PCSR-only build).

- **Page shows "Connecting to the index"**: wait 10 seconds; if it persists, restart `make web`.
- **Port 8000 busy**: `python app/web_server.py --port 9000`.
- **Ollama not listed under "Answer by"**: it is not running; use the E20/E25 figures (scene 10).
- **Anything else**: show the screenshot of that scene, or the command line:

```
python app/cli.py "punishment for murder" --date 2025-01-03 --agent --answer
python app/cli.py "" --drift ipc:124a
```